

Program : Diploma Robotics Process Automation	
Course Code : 3309	Course Title: Introduction to Robotic Process Automation Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide hands on experiments with the basic concepts of Robotic Process Automation
- To learn how to install UiPath Studio Community Edition.
- To Develop Bots using UiPath.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic functions and features of Computer, Operating system and Internet applications.		Introduction to IT systems Lab	1
Basic programming skills in C language.		Introduction to Computation and Programming	2

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Installation and Familiarization of UiPath Studio Community Edition tool	7	Applying
CO2	Develop Bots using types of variables and arguments	8	Applying

CO3	Implement Bots using control flow statements	10	Applying
CO4	Develop Bots using Basic and Desktop recording	17	Applying
	Series Test	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3	3			
CO2	3	3	3	3			
CO3	3	3	3	3			
CO4	3	3	3	3			

3 – Strongly mapped

2 – Moderately mapped

1 – Weakly mapped

Course Outline:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Installation and Familiarization of UiPath Studio Community Edition tool		
M1.01	Installation of UiPath Studio Community Edition	1	Applying
M1.02	Familiarization of UiPath components- Activities and Sequence	2	Applying
M1.03	Familiarization of the UiPath Interface components- The Ribbon, The Universal Search Bar, The Designer Panel	2	Applying
M1.04	Familiarization of the UiPath Interface components- The Activities Panel, The Library Panel, The Project Panel, The Properties Panel, The Outline Panel	2	Applying

CO2	Develop Bots using types of variables and arguments		
M2.01	Create a simple bot which will greet you with the name i.e. “Welcome to UiPath Studio”	1	Applying
M2.02	Create a welcome message using variable	1	Applying
M2.03	Develop an automation using text variables	1	Applying
M2.04	Implement an automation using True or False variables.	1	Applying
M2.05	Develop an automation using integer variables	1	Applying
M2.06	Implement an automation using array of variables	1	Applying
M2.07	Create an automation using date and time variable	1	Applying
M2.08	Develop an automation using arguments	1	Applying
	Series Test 1	1.5	
CO3	Implement Bots using control flow statements.		
M3.01	Create an automation using delay activity	2	Applying
M3.02	Develop a bot using if activity – Example: build a bot that prints a message based on whether a cell has a value or is empty	2	Applying
M3.03	Develop a bot using switch activity – Build a bot which asks the user to input some numbers and perform simple arithmetic operations like addition, subtraction, multiplication and division as per users wish and displays the result as an output message.	2	Applying
M3.04	Implement an automation using do-while, while activity	2	Applying
M3.05	Create a bot using for each activity	2	Applying
CO4	Develop Bots using Basic and Desktop recording.		
M4.01	Create a simple bot to automate a manual task which includes a sequence of mouse movements and keyboards	2	Applying
M4.02	Implement a simple bot to automate a manual task which opening a desktop application and perform some activity like opening notepad and typing some data into it and saving the text file.	2	Applying

M4.03	Create a simple bot to open a browser and visit a website/url of interest and then close the browser	2	Applying
M4.04	Develop a simple bot to read the contents of an excel sheet and print them in a message box.	3	Applying
M4.05	Create Bots that performs simple excel data manipulation	3	Applying
M4.06	Create a Screen scrapping Bot	5	Applying
	Series Test 2	1.5	

**** - Micro Project**

Students are expected to do a micro project using UiPath Studio edition tool during the course for the purpose of continuous evaluation. This experiment shall be included in the bona-fide record.

Example:

A simple screen scrapping robot

Excel data sorting etc.

Text / Reference:

T/R	Book Title/Author
T1	Alok Mani Tripathi, “Learning Robotic Process Automation”, Packt Publishing, 2018
R1	Frank Casale , Rebecca Dilla, Heidi Jaynes , Lauren Livingston, “Introduction to Robotic Process Automation: a Primer”, Institute of Robotic Process Automation, 1st Edition 2015.
R2	Richard Murdoch, Robotic Process Automation: Guide To Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant”, Independently Published, 1st Edition 2018.

Online Resources:

Sl.No	Website Link
1	https://docs.uipath.com/studio/docs

2	https://dev.tutorialspoint.com/uiopath/uiopath_tutorial.pdf
3	https://www.udemy.com/course/robotic-process-automation/

List of Sample Experiments

Write a python program to find the sum of 2 numbers.

Write a python program to find the area of a circle.

Write a python program to find the area and perimeter of a rectangle.

Write a python program to swap 2 numbers using temporary variable.

Write a python program to swap 2 numbers without using temporary variable.

Write a python program to convert celcius to fahrenheit.

Write a python program to calculate simple interest.

Write a python program to check whether the given number is divisible by 2 or 3.

Write a python program to find the largest of 2 numbers.

Write a python program to find the largest of 3 numbers.

Write a python program to find whether the given number is even or not.

Write a python program to find whether the given number is positive or negative.

Write a python program to find the largest of n numbers.

Write a python program to find the sum of n numbers.

1 Write a python program to find the factorial of a number.

Write a python program to display fibonacci series.

Write a python program to find the sum of a sequence of numbers between a lower bound and upper bound.

Write a python program to find factorial using function.

Write a python program to reverse a number using function.

Write a python program to find the sum of digits of a number using function.

Write a python program to display the multiplication table of a given number.

Write a python program to find the factorial of a number using recursion

Write a python program to check whether a given string is palindrome or not.

Write a python program to count the number of times each character appears in a string.

Write a python program to compare two strings.

Write a python program which accepts a string, splits it and outputs the list of words in uppercase.

Write a python program which accepts a list of numbers and outputs their squares.

Write a python program to swap 2 numbers using tuples and function.

Write a python program to generate a list of random values using function.

Write a python program to find the median of a set of numbers using list.

Write a python program to convert hex to binary using dictionary.

Write a python program to generate a histogram of the letters in a string using dictionary.

Write a python program to copy contents of one file to another omitting any lines that begin with

Write a python program to open and write your roll no an list of marks to a file using pickling.

Micro project implementation.